Digital Forensic Investigation Lifecycle using Python

Python Program:

```
import datetime

def investigation_report(case_id, evidence_list):
  stages = {
    "Identification": f"Incident reported for case {case_id}.
Devices/logs identified for review.",
    "Collection": f"{len(evidence_list)} item(s) collected: {',
'.join(evidence_list)}",
    "Preservation": "All items hashed (SHA-256) and stored in a
write-protected evidence folder.",
    "Analysis": "Log files and file metadata examined for indicators of
compromise.",
    "Reporting": "Findings compiled into a structured forensic report for
legal review."
  }

  print(f"=== Investigation Lifecycle: Case {case_id} ===")
  print(f"Generated: {datetime.datetime.now()}\n")

  for stage, detail in stages.items():
    print(f"[{stage}]")
    print(f" {detail}\n")

investigation_report("CASE-2026-014",
["laptop_disk_image.dd","router_traffic.pcap","email_headers.txt"])
```

Sample Output:

=== Investigation Lifecycle: Case CASE-2026-014 ===
Generated: 2026-07-16 05:43:16.508220

[Identification] Incident reported for case CASE-2026-014.
[Collection] 3 item(s) collected: laptop_disk_image.dd, router_traffic.pcap, email_headers.txt.
[Preservation] All items hashed (SHA-256) and stored safely.
[Analysis] Log files and metadata examined.
[Reporting] Findings compiled into a forensic report.

Explanation:

This program demonstrates the basic stages of a digital forensic investigation. It generates an investigation report by displaying the five main phases: Identification, Collection, Preservation, Analysis, and Reporting. The program also records the report generation time and lists the collected evidence, providing a simple example of the digital forensic investigation lifecycle used in cybersecurity and digital forensics.